

Suggested Solution 2022 final exam

Problem 1

Tasks:

- i) [8%] Describe the Δu formulation method for obtaining integral action in detail. Show clearly all the mathematical details of how the augmented model is derived.

Suggested Solution:

The term Δu is defined as,

$$\Delta u_k = u_k - u_{k-1} \quad (s1)$$

The linear system's model is given by,

$$x_{k+1} = Ax_k + Bu_k \quad (s2)$$

$$y_k = Cx_k \quad (s3)$$

The offset due to mismatch between linear model (with fixed operating point) and the nonlinear model can be considered to be constant or very slowly varying. To get rid of these offsets, we can define $\Delta x_k = (x_k - x_{k-1})$ or with one increment in k index as $\Delta x_{k+1} = (x_{k+1} - x_k)$. Then we have,

$$x_{k+1} - x_k = Ax_k + Bu_k - Ax_{k-1} - Bu_{k-1} \quad (s4)$$

$$\Delta x_{k+1} = A(x_k - x_{k-1}) + B(u_k - u_{k-1}) \quad (s5)$$

$$\Delta x_{k+1} = A\Delta x_k + B\Delta u_k \quad (s6)$$

Similarly, for the output equation we have,

$$y_k - y_{k-1} = Cx_k - Cx_{k-1} \quad (s7)$$

$$y_k - y_{k-1} = C(x_k - x_{k-1}) \quad (s8)$$

$$y_k = y_{k-1} + C\Delta x_k \quad (s9)$$

Now we can augment Equation s6 and Equation s9 as,

$$\underbrace{\begin{bmatrix} \Delta x_{k+1} \\ y_k \end{bmatrix}}_{\tilde{x}_{k+1}} = \underbrace{\begin{bmatrix} A & 0 \\ C & I \end{bmatrix}}_{\tilde{A}} \underbrace{\begin{bmatrix} \Delta x_k \\ y_{k-1} \end{bmatrix}}_{\tilde{x}_k} + \underbrace{\begin{bmatrix} B \\ 0 \end{bmatrix}}_{\tilde{B}} \Delta u_k \quad (s10)$$

$$y_k = \underbrace{\begin{bmatrix} C & I \end{bmatrix}}_{\tilde{C}} \underbrace{\begin{bmatrix} \Delta x_k \\ y_{k-1} \end{bmatrix}}_{\tilde{x}_k} \quad (s11)$$

The augmented model is in a standard linear space form as,

$$\tilde{x}_{k+1} = \tilde{A}\tilde{x}_k + \tilde{B}\Delta u_k \quad (s12)$$

$$y_k = \tilde{C}\tilde{x}_k \quad (s13)$$

- ii) [8%] Now use the augmented model to write down the optimal control problem formulation (the objective function and the constraints) for designing a linear MPC having the integral action feature.

Suggested solution:

The LQ optimal control should be constructed with the augmented model of Equation s12 and s13. Since the augmented linear model of the process now contains Δu_k instead of u_k , the formulation of the

performance criteria (i.e. the objective function) for the optimal control problem and the constraints should also be modified to contain Δu_k as,

$$\min_{(\Delta u_k)} J = \frac{1}{2} \sum_{k=1}^N e_k^T Q_k e_k + \Delta u_{k-1}^T P_{k-1} \Delta u_{k-1} \quad (s14)$$

subject to,

$$\tilde{x}_{k+1} = \tilde{A} \tilde{x}_k + \tilde{B} \Delta u_k \quad \text{with } \tilde{x}_k \in \mathbb{R}^{(n_x+n_y) \times 1}, \Delta u_k \in \mathbb{R}^{n_u \times 1} \text{ and } \tilde{x}_0 \text{ known} \quad (s15)$$

$$y_k = \tilde{C} \tilde{x}_k \quad \text{with } y_k \in \mathbb{R}^{n_y \times 1} \quad (s16)$$

$$e_k = r_k - y_k \quad (s17)$$

$$\Delta u_L \leq \Delta u_k \leq \Delta u_U \quad (s18)$$

$$u_L \leq u_k \leq u_U \quad (s19)$$

If any of the augmented states $\tilde{x}_k = \begin{bmatrix} \Delta x_k \\ y_{k-1} \end{bmatrix}$ are not measurable then they should be estimated using state estimators like the Kalman filter. Here the variables to be optimized are the deviation of the control input or rate of change of control input Δu_k . However, when applying the control input to the system, $u_k = \Delta u_k + u_{k-1}$ should be used.

- iii) Now, in order to solve your MPC problem with integral action, you should convert the optimal control problem of task (ii) into a standard QP problem given by equation 4 in an efficient way (say using kronecker product formulation).

$$\begin{aligned} \min_z \quad & \frac{1}{2} z^T H z + c^T z, \\ \text{s.t.} \quad & A_e z = b_e, \\ & z_L \leq z \leq z_H. \end{aligned} \quad (4)$$

The vector of unknowns, z , must include the following elements from the **augmented** model in their respective places: [control inputs, states, error and measurement].

Answer the following for task (iii):

- (a) [10%] Find the expression for matrix H and vector c .

Suggested solution:

Let us for example define the vector of unknowns z as follows,

$$z = \begin{bmatrix} \Delta u \\ \tilde{x} \\ e \\ y \end{bmatrix} \quad \text{i.e.} \quad z^T = (\Delta u^T, \tilde{x}^T, e^T, y^T)$$

Based on our choice, the total number of unknowns (n_z) in vector z for the whole prediction horizon length (N) is,

$$n_z = N (n_u + n_x + n_y + n_y)$$

If we expand the objective function J of equation (p8) for $k = 1$ to $k = N$ we will get,

$$\begin{aligned} J = \frac{1}{2} [& e_1^T Q_1 e_1 + e_2^T Q_2 e_2 + \dots + e_N^T Q_N e_N + \Delta u_0^T P_0 \Delta u_0 + \Delta u_1^T P_1 \Delta u_1 + \dots \\ & + \Delta u_{N-1}^T P_{N-1} \Delta u_{N-1}] \end{aligned} \quad (p13)$$

Now with our choice of vector z , the standard quadratic objective function of equation 4 can be written as,

$$J = \frac{1}{2} z^T H z + c^T z$$

$$J = \frac{1}{2} \underbrace{\begin{bmatrix} \Delta u \\ \tilde{x} \\ y \end{bmatrix}^T}_{z^T} \underbrace{\begin{bmatrix} H_{11} & 0 & 0 & 0 \\ 0 & H_{22} & 0 & 0 \\ 0 & 0 & H_{33} & 0 \\ 0 & 0 & 0 & H_{44} \end{bmatrix}}_H \underbrace{\begin{bmatrix} \Delta u \\ \tilde{x} \\ e \\ y \end{bmatrix}}_z + \underbrace{\begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}^T}_{c^T} \underbrace{\begin{bmatrix} \Delta u \\ \tilde{x} \\ e \\ y \end{bmatrix}}_z \quad (\text{p14})$$

Now if we multiply the matrices in equation (p14) we get,

$$J = \frac{1}{2} [\Delta u^T H_{11} \Delta u + \tilde{x}^T H_{22} \tilde{x} + e^T H_{33} e + y^T H_{44} y] + c_1^T \Delta u + c_2^T \tilde{x} + c_3^T e + c_4^T y \quad (\text{p15})$$

Comparing (p13) with (p15),

$$\Delta u^T H_{11} \Delta u = \Delta u_0^T P_0 \Delta u_0 + \Delta u_1^T P_1 \Delta u_1 + \dots + \Delta u_{N-1}^T P_{N-1} \Delta u_{N-1}$$

In matrix form,

$$\Delta u^T H_{11} \Delta u = \begin{bmatrix} \Delta u_0 \\ \Delta u_1 \\ \vdots \\ \Delta u_{N-1} \end{bmatrix}^T \begin{bmatrix} P_0 & 0 & \dots & 0 \\ 0 & P_1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & P_{N-1} \end{bmatrix} \begin{bmatrix} \Delta u_0 \\ \Delta u_1 \\ \vdots \\ \Delta u_{N-1} \end{bmatrix} \quad (\text{p16})$$

Comparing both sides of equation (p16) we finally can write,

$$H_{11} = \begin{bmatrix} P_0 & 0 & \dots & 0 \\ 0 & P_1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & P_{N-1} \end{bmatrix}$$

If, $P_0 = P_1 = \dots = P_{N-1} = P$ with $P \in \mathbb{R}^{n_u \times n_u}$, then we have

$$H_{11} = \begin{bmatrix} P & 0 & \dots & 0 \\ 0 & P & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & P \end{bmatrix} = I_N \otimes P$$

Note:

I_N = Identity matrix of size 'N'

$\otimes$ = Kronecker product

Again comparing (p13) with (p15) we see that there is no ' $\tilde{x}$ ' term in equation (p15). So we have,

$$H_{22} = \begin{bmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix} = \mathbf{0}_{N.n_x \times N.n_x} = I_N \otimes \mathbf{0}_{n_x \times n_x}$$

Similarly, for the error term, comparing (p13) with (p15) we have,

$$e^T H_{33} e = e_1^T Q_1 e_1 + e_2^T Q_2 e_2 + \dots + e_N^T Q_N e_N$$

In matrix form,

$$e^T H_{33} e = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_N \end{bmatrix}^T \begin{bmatrix} Q_1 & 0 & \dots & 0 \\ 0 & Q_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & Q_N \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_N \end{bmatrix} \quad (\text{p17})$$

Comparing both sides of equation p17 we get,

$$H_{33} = \begin{bmatrix} Q_1 & 0 & \dots & 0 \\ 0 & Q_2 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & & \dots & Q_N \end{bmatrix}$$

If, $Q_1 = Q_2 = \dots = Q_N = Q$ with $Q \in R^{n_y \times n_y}$, we get,

$$H_{33} = \begin{bmatrix} Q & 0 & \dots & 0 \\ 0 & Q & & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & Q \end{bmatrix} = I_N \otimes Q$$

Again comparing (p13) with (p15) we see that there is no 'y' term in equation (p15). So we have,

$$H_{44} = \begin{bmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix} = \mathbf{0}_{N.n_y \times N.n_y} = I_N \otimes \mathbf{0}_{n_y \times n_y}$$

Then the H matrix of the standard quadratic objective function can be written as,

$$H = \text{blkdiag}(H_{11}, H_{22}, H_{33}, H_{44}) \\ = \text{blkdiag}(I_N \otimes P, \mathbf{0}_{N.n_x \times N.n_x}, I_N \otimes Q, \mathbf{0}_{N.n_y \times N.n_y})$$

In addition, we can clearly see that in equation (p15) we do not have any linear term. Then comparing equations (p13) with (p15) we get,

$$c = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} 0_{N.n_u} \\ 0_{N.n_x} \\ 0_{N.n_y} \\ 0_{N.n_y} \end{bmatrix} = \mathbf{0}_{(n_z \times 1)} \quad n_z = \text{no. of total unknowns}$$

(b) [12%] Find the expression for matrix A_e and vector b_e .

Suggested solution:

Now let us express the equality constraints of the LQ optimal control problem given by equation (s15) - (s17) in the standard QP form $A_e z = b_e$ given by equation 4. Let us first organize the matrix A_e and vector b_e as follows,

$$\underbrace{\begin{bmatrix} A_{e,1\Delta u} & A_{e,1\tilde{x}} & A_{e,1e} & A_{e,1y} \\ A_{e,2\Delta u} & A_{e,2\tilde{x}} & A_{e,2e} & A_{e,2y} \\ A_{e,3\Delta u} & A_{e,3\tilde{x}} & A_{e,3e} & A_{e,3y} \end{bmatrix}}_{A_e} \underbrace{\begin{bmatrix} u \\ \tilde{x} \\ e \\ y \end{bmatrix}}_{b_e} = \underbrace{\begin{bmatrix} b_{e,1} \\ b_{e,2} \\ b_{e,3} \end{bmatrix}}_{b_e} \quad (\text{p18})$$

Each row of A_e in equation (p18) corresponds to each equality constraint of equations (s15) - (s17). Each column of A_e represents an element of the unknown vector z . Let us first consider the equality constraint given by equation (s15)

$$\tilde{x}_{k+1} = \tilde{A}\tilde{x}_k + \tilde{B}\Delta u_k \\ \tilde{x}_k - \tilde{A}\tilde{x}_{k-1} - \tilde{B}\Delta u_{k-1} = 0$$

We should obey the constraints for the whole prediction horizon length. Since we consider a prediction horizon from $k = 1$ to $k = N$ we have,

$$\begin{aligned} \tilde{x}_1 - \tilde{A}\tilde{x}_0 - \tilde{B}\Delta u_0 &\rightarrow \tilde{x}_1 - \tilde{B}\Delta u_0 = \tilde{A}\tilde{x}_0 & \text{for } k = 1 \\ \tilde{x}_2 - \tilde{A}\tilde{x}_1 - \tilde{B}\Delta u_1 &= 0 & \text{for } k = 2 \end{aligned} \quad (\text{p19})$$

$$\begin{matrix} \vdots \\ \tilde{x}_N - \tilde{A}\tilde{x}_{N-1} - \tilde{B}\Delta u_{N-1} = 0 \\ \vdots \end{matrix} \quad \text{for } k = N$$

Arranging the set of equations (p19) into a matrix form we get the first row of A_e in equation (p18) as,

$$\begin{bmatrix} \underbrace{\begin{bmatrix} -\tilde{B} & 0 & 0 & \dots & 0 \\ 0 & -\tilde{B} & 0 & \dots & 0 \\ 0 & 0 & -\tilde{B} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & \dots & -\tilde{B} \end{bmatrix}}_{A_{e,1\Delta u}} & \vdots & \underbrace{\begin{bmatrix} I & 0 & 0 & \dots & 0 \\ -\tilde{A} & I & 0 & \dots & 0 \\ 0 & -\tilde{A} & I & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & -\tilde{A} & I \end{bmatrix}}_{A_{e,1\tilde{x}}} & \vdots & \underbrace{\begin{bmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & & \dots & \\ 0 & 0 & & \ddots & \\ \vdots & \vdots & & \vdots & \\ 0 & 0 & \dots & \dots & 0 \end{bmatrix}}_{A_{e,1e}} & \vdots & \underbrace{\begin{bmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & & \dots & \\ 0 & 0 & & \ddots & \\ \vdots & \vdots & & \vdots & \\ 0 & 0 & \dots & \dots & 0 \end{bmatrix}}_{A_{e,1y}} \end{bmatrix}_{(N.n_x \times n_z)} \begin{bmatrix} \Delta u_0 \\ \Delta u_1 \\ \vdots \\ \Delta u_{N-1} \\ \tilde{x}_1 \\ \tilde{x}_2 \\ \vdots \\ \tilde{x}_N \\ e \\ y \end{bmatrix}_{(n_z \times 1)} = \underbrace{\begin{bmatrix} \tilde{A}x_0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}}_{b_{e,1}}$$

So, we have from the first linear equality constraint,

$$A_{e,1\Delta u} = \begin{bmatrix} -\tilde{B} & 0 & \dots & 0 \\ 0 & -\tilde{B} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & -\tilde{B} \end{bmatrix} = -I_N \otimes \tilde{B}$$

$$A_{e,1\tilde{x}} = \begin{bmatrix} I & 0 & 0 & \dots & 0 \\ -\tilde{A} & I & 0 & \dots & 0 \\ 0 & -\tilde{A} & I & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & -\tilde{A} & I \end{bmatrix} = I_{N.n_x} - (I_{N,-1} \otimes \tilde{A})$$

Similarly,

$$A_{e,1e} = \mathbf{0}_{(N.n_x \times N.n_y)}$$

$$A_{e,1y} = \mathbf{0}_{(N.n_x \times N.n_y)}$$

$$\text{and } b_{e,1} = \begin{bmatrix} \tilde{A}x_0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{(N.n_x \times 1)}$$

Now considering the second equality constraint of equation (p10) we get,

$$y - \tilde{C}\tilde{x}_k = 0$$

This equality constraint must also be satisfied over the whole prediction horizon length. Since we consider a prediction horizon from $k = 1$ to $k = N$ we get,

$$\begin{aligned} y_1 - \tilde{C}\tilde{x}_1 &= 0_{(n_y \times 1)} & \text{for } k &= 1 \\ y_2 - \tilde{C}\tilde{x}_2 &= 0_{(n_y \times 1)} & \text{for } k &= 2 \\ \vdots & & \vdots & \\ y_N - \tilde{C}\tilde{x}_N &= 0_{(n_y \times 1)} & \text{for } k &= N \end{aligned} \tag{p20}$$

Arranging the set of equations (p20) obtained from the second equality constraints into a matrix form we get the second row of A_e in equation (p18) as,

$$\begin{bmatrix} \underbrace{\begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{bmatrix}}_{A_{e,2\Delta u}} & \vdots & \underbrace{\begin{bmatrix} -\tilde{C} & 0 & \cdots & 0 \\ 0 & -\tilde{C} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & -\tilde{C} \end{bmatrix}}_{A_{e,2\tilde{x}}} & \vdots & \underbrace{\begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{bmatrix}}_{A_{e,2e}} & \vdots & \underbrace{\begin{bmatrix} I & 0 & \cdots & 0 \\ 0 & I & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & I \end{bmatrix}}_{A_{e,2y}} \end{bmatrix}_{(N.n_y \times n_z)} \begin{bmatrix} \Delta u \\ \tilde{x}_1 \\ \tilde{x}_2 \\ \vdots \\ \tilde{x}_N \\ e \\ y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}_{(n_z \times 1)} = \begin{bmatrix} 0_{n_y \times 1} \\ 0_{n_y \times 1} \\ \vdots \\ \vdots \\ \vdots \\ 0_{n_y \times 1} \end{bmatrix}_{\underbrace{b_{e,2}}_{(n_z \times 1)}}$$

So we have,

$$A_{e,2\Delta u} = \mathbf{0}_{(N.n_y \times N.n_u)}$$

$$A_{e,2\tilde{x}} = \begin{bmatrix} -\tilde{C} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & -\tilde{C} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & -\tilde{C} \end{bmatrix} = -I_N \otimes \tilde{C}$$

$$A_{e,2e} = \mathbf{0}_{(N.n_y \times N.n_y)}$$

$$A_{e,2y} = \begin{bmatrix} I & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & I & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & I \end{bmatrix} = I_{N.n_y}$$

$$b_{e,2} = \mathbf{0}_{(N.n_y \times 1)}$$

Finally, from the last equality constraint of equation (p10) we get,

$$e_k + y_k = r_k$$

This equality constraint must also be satisfied over the whole prediction horizon length. Since we consider a prediction horizon from $k = 1$ to $k = N$ we get,

$$\begin{aligned} e_1 + y_1 &= r_1 & \text{for } k &= 1 \\ e_2 + y_2 &= r_2 & \text{for } k &= 2 \\ \vdots & \vdots & \vdots & \\ e_N + y_N &= r_N & \text{for } k &= N \end{aligned} \tag{p21}$$

Arranging the set of equations (p21) obtained from the third equality constraints in a matrix form we get the third row of A_e in equation (p18) as,

$$\begin{bmatrix} \underbrace{\begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{bmatrix}}_{A_{e,3\Delta u}} & \vdots & \underbrace{\begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{bmatrix}}_{A_{e,3\tilde{x}}} & \vdots & \underbrace{\begin{bmatrix} I & 0 & \cdots & 0 \\ 0 & I & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & I \end{bmatrix}}_{A_{e,3e}} & \vdots & \underbrace{\begin{bmatrix} I & 0 & \cdots & 0 \\ 0 & I & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & I \end{bmatrix}}_{A_{e,3y}} \end{bmatrix}_{(N.n_y \times n_z)} \begin{bmatrix} u \\ \tilde{x} \\ e_1 \\ e_2 \\ \vdots \\ e_N \\ y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}_{(n_z \times 1)} = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ \vdots \\ \vdots \\ r_N \end{bmatrix}_{\underbrace{b_{e,3}}_{(n_z \times 1)}}$$

So we have,

$$A_{e,3\Delta u} = \mathbf{0}_{(N.n_y \times N.n_u)}$$

$$A_{e,3\tilde{x}} = \mathbf{0}_{(N.n_y \times N.n_x)}$$

$$A_{e,3e} = \begin{bmatrix} I & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & I & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & I \end{bmatrix} = I_N \otimes I_{n_y} = I_{N.n_y}$$

CANDIDATES MUST THEMSELVES CHECK THAT ALL ASSIGNMENTS AND ATTACHMENTS ARE IN ORDER.

$$A_{e,3y} = \begin{bmatrix} I & 0 & \dots & 0 \\ 0 & I & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & I \end{bmatrix} = I_{N.n_y}$$

$$b_{e,3} = \begin{bmatrix} r_1 \\ \vdots \\ r_N \end{bmatrix}_{(N.n_y \times 1)}$$

Finally, all the three equality constraints can be written in the standard form of $A_e z = b_e$ where,

$$A_e = \begin{bmatrix} -I_N \otimes \tilde{B} & I_{N.n_x} - (I_{N,-1} \otimes \tilde{A}) & 0_{N.n_x \times N.n_y} & 0_{N.n_x \times N.n_y} \\ 0_{N.n_y \times N.n_u} & -I_N \otimes \tilde{C} & 0_{N.n_y \times N.n_y} & I_{N.n_y} \\ 0_{N.n_y \times N.n_u} & 0_{N.n_y \times N.n_x} & I_{N.n_y} & I_{N.n_y} \end{bmatrix}$$

and

$$b_e = \begin{bmatrix} b_{e,1} \\ b_{e,2} \\ b_{e,3} \end{bmatrix} = \begin{bmatrix} \tilde{A}\tilde{x}_0 \\ 0 \\ \vdots \\ 0 \\ \mathbf{0}_{(N.n_y \times 1)} \\ r_1 \\ \vdots \\ r_N \end{bmatrix}$$

Problem 2

Tasks

[8%] Show with mathematical descriptions as to how terminal costs and terminal constraints can be utilized for improving the stability of a finite horizon MPC.

Suggested solution:

A finite horizon MPC cannot guarantee stability even if a feasible solution exists at every time step. Let us write a finite horizon MPC in general as,

$$(A) \left\{ \begin{array}{l} \min_{(u_0, u_1, \dots, u_{N-1})} \quad J = \sum_{k=0}^{N-1} q(x_k, u_k) \\ \text{s.t.} \quad x_{k+1} = f(x_k, u_k) \quad , \quad k = 0, 1, 2, \dots, N-1 \rightarrow \text{nonlinear process model,} \quad x_0 \text{ known} \\ y_k = g(x_k, u_k) \quad , \quad k = 0, 1, 2, \dots, N-1 \rightarrow \text{measurement equations} \\ h_i(x) - b_i = 0 \quad , \quad i = 1, 2, \dots, m \rightarrow \text{nonlinear equality constraints} \\ I_j(x) - c_j \leq 0 \quad , \quad j = 1, 2, \dots, r \rightarrow \text{nonlinear inequality constraints} \end{array} \right.$$

Here, $q(x_k, u_k)$ is any linear or nonlinear objective. The stability of the finite horizon constrained optimal control problem of (A) can be improved by making the horizon longer. However, to ensure stability, the prediction horizon can be extended to infinity in principle, i.e. by considering an infinite horizon MPC. So, what we would like the MPC to be is given by problem (B),

$$(B) \left\{ \begin{array}{l} \text{Min} \\ (u_0, u_1, \dots, \dots, \infty) \quad J = \sum_{k=0}^{\infty} q(x_k, u_k) \\ \text{s.t.} \quad x_{k+1} = f(x_k, u_k) \quad , \quad k = 0, 1, 2, \dots, \infty, x_0 \text{ known} \\ \quad \quad y_k = g(x_k, u_k) \quad , \quad k = 0, 1, 2, \dots, \infty \\ \quad \quad h_i(x) - b_i = 0 \quad , \quad i = 1, 2, \dots, m \\ \quad \quad I_j(x) - c_j \leq 0 \quad , \quad j = 1, 2, \dots, r \end{array} \right.$$

But infinite horizon constrained optimal control problem of (B) is infinite dimensional optimization problem. It cannot be computed or solved using conventional method because there are infinite number of variables to optimize. At the same time, the objective function should also be bounded i.e. $J < \infty$.

The idea is to split the infinite horizon of problem (B) into two parts:

$$\begin{array}{ll} 0, 1, \dots, N-1, & : \text{part 1} \\ N, N+1, \dots, \infty & : \text{part 2} \end{array}$$

Splitting is done in a way that the first part $(0, 1, \dots, N-1)$ will represent a finite horizon control problem. Then we have,

$$(C) \left\{ \begin{array}{l} \text{min} \\ (u_0, u_1, \dots, \dots, u_{\infty}) \quad J = \sum_{k=0}^{N-1} q(x_k, u_k) + \sum_{k=N}^{\infty} q(x_k, u_k) \\ \text{s.t.} \quad x_{k+1} = f(x_k, u_k) \quad , \quad k = 0, 1, 2, \dots, \infty \\ \quad \quad y_k = g(x_k, u_k) \quad , \quad k = 0, 1, 2, \dots, \infty \\ \quad \quad h_i(x) - b_i = 0 \quad , \quad i = 1, 2, \dots, m \\ \quad \quad I_j(x) - c_j \leq 0 \quad , \quad j = 1, 2, \dots, r \\ \quad \quad x_0 \text{ known} \end{array} \right.$$

N should be large enough to drive the system/process to a state $x_N \in \chi_f$ where χ_f is a feasible set. The objective function in the infinite horizon interval (the second part) can for simplicity be written in quadratic form as

$$\sum_{k=N}^{\infty} q(x_k, u_k) = p(x_N) = \frac{1}{2} x_N^T S x_N$$

where S is the weighting matrix for x_N . The term $p(x_N)$ is also referred to as terminal cost or terminal objective function. The term $x_N \in \chi_f$ will give rise to terminal constraints. Then problem (C) which is infinite horizon constrained problem can be reformulated as finite horizon constrained problem or finite dimensional control problem (D) as,

$$(D) \left\{ \begin{array}{l} \text{Min} \\ (u_0, u_1, \dots, \dots, u_{N-1}) \quad J = \sum_{k=0}^{N-1} q(x_k, u_k) + p(x_N) \\ \text{s.t.} \quad x_{k+1} = f(x_k, u_k) \quad , \quad k = 0, 1, 2, \dots, N-1 \\ \quad \quad y_k = g(x_k, u_k) \quad , \quad k = 0, 1, 2, \dots, N-1 \\ \quad \quad h_i(x) - b_i = 0 \quad , \quad i = 1, 2, \dots, m \\ \quad \quad I_j(x) - c_j \leq 0 \quad , \quad j = 1, 2, \dots, r \\ \quad \quad x_0 \text{ known} \\ \quad \quad x_N \in \chi_f \quad \rightarrow \text{terminal constraint} \end{array} \right.$$

Note that in the optimal control problem (D), $U \in [u_0, u_1, \dots, u_{N-1}]$ i.e. we have finite number of variables to optimize. By introducing terminal cost and terminal constraint, stability is ensured (of course for a proper choice of N) provided that the problem is feasible at all times.

Note: For a proper choice of N , it can be proven that the terminal objective function $p(x_N)$ becomes a Lyapunov function for the whole horizon that decays toward the origin i.e. stability is guaranteed. The proof

involves a good choice of N and the generation of χ_f is a difficult task (especially for nonlinear system) and it is out of the scope of the course. Generally, in practice, terminal sets are not used i.e. simply neglected as long as there is feedback in the system.

Problem 3

Tasks

- (i) [7%] What is the difference between a state feedback MPC and an output feedback MPC. Draw necessary signal flow diagrams (block diagrams) to illustrate your reasoning.

Suggested solution:

A state feedback MPC as the name implies is a controller where the states are directly fed back to the controller. This is only possible if all the states of the process being controlled are available or measurable i.e. full state information is a necessity. This makes the state feedback MPC an ideal case. The block diagram in figure (b) illustrates the state feedback MPC.

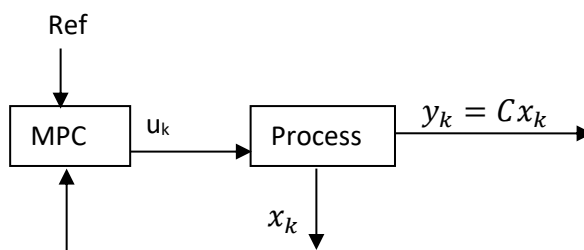


Figure (b): State feedback MPC

However, in practice, it may not always be possible to measure all the state of the system. In such cases, the states of the system should be "ESTIMATED". Estimation of the states can be performed by utilizing the available measurements. So in this case, the measurements (output) of the process is fed to an estimator (say a Kalman filter) which instead estimates the states. These estimated states are then fed to the MPC as shown in the block diagram in Figure (c).

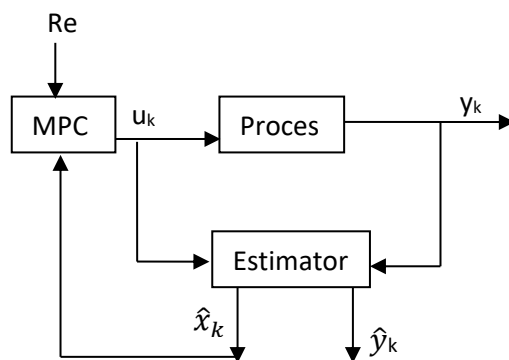


Figure (c): Output feedback MPC with a state estimator

- (ii) [8%] Explain the weighted sum method for solving a multi objective optimization problem with necessary diagram. Also write down its disadvantages if there are any.

Suggested solution:

With a multi-objective optimization there are more than one, let's say there are k number of objectives functions to be minimized simultaneously. Let us also assume that these objective functions are conflicting to each other. With the weighted sum method, each objectives are assigned a weighting factor $w_i > 0$ and then these objective functions are summed together as shown below,

$$\underset{u}{\text{Min}} \quad J(x, u) = \sum_{i=1}^k w_i F_i(x, u)$$

s. t

< some constraints >

Here $F_i(x, u)$ are individual objective functions. To explain the concept behind weighted sum method, let us take an example of a multi-objective optimization problem with two conflicting objectives such that they both are to be minimized as shown in figure (d).

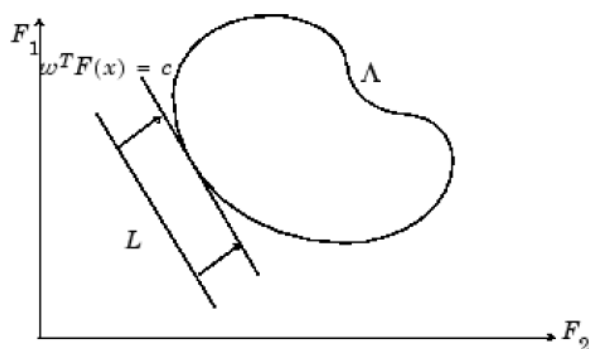


Figure (d): Geometric representation of weighted sum method

The slope of the line L depends on the choice of the weighting factor. For one choice of w_i 's, the slope of the line is fixed. For another choice of w_i 's the slope of the line can be different. When solving the optimization problem, at each iteration, the objective function $J(x, u)$ becomes better and better, and the line slides towards the feasible region denoted by symbol Λ in figure (d). As the iteration proceeds, the point where the line L touches the feasible region is the solution of the problem.

The disadvantage of this method is that it can be difficult to assign proper weights, and it is usually based on experience or trial and error. For non-convex problem, certain non-dominated solutions are not accessible as shown in figure (e). Any point lying between A and B in figure (e) is not accessible by this method.

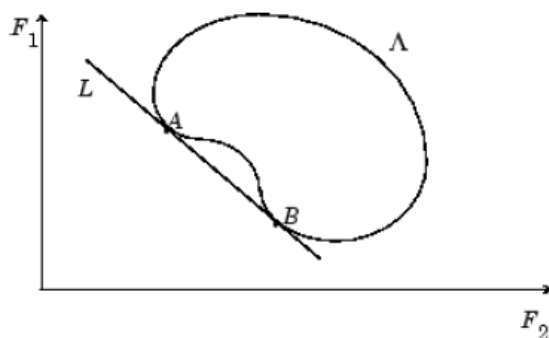


Figure (e): Illustration for showing disadvantage of weighted sum method.